SAP Business Data Cloud (Data Analytics Engineer) Record-to-Report (R2R) - Financial Close Analytics Pipeline

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1. Problem Statement

Finance controllers in SAP S/4HANA environments face a critical challenge during month-end and year-end close, GL data across accounts, cost centres, and profit centres cannot be consolidated in real time using BEx Analyzer or Report Painter. Decision-makers need a unified view of P&L, trial balance, cost centre variance, AR ageing, and DSO with period-over-period comparison and automated data product publishing. This project delivers an end-to-end Sap Record to Report Analytics pipeline covering all six R2R sub-processes named as GL posting, P&L close, trial balance, cost centre allocation, profit centre reporting, and AR ageing.

2. Solution and Features

Data Ingestion: Simulates 600 SAP GL journal entries (BKPF/BSEG/FAGLFLEXT) and 200 AR invoices across 6 cost centres, 4 profit centres, 8 GL accounts, and 12 fiscal periods (P01–P12/2024) as would arrive via SDI/SLT replication in real SAP Business Data Cloud.

BW/4HANA Bridge - 6 Financial OLAP queries

Query	BW Equivalent	Output
Monthly P&L	CompositeProvider on FAGLFLEXT	Revenue, Expense, Net Profit, Margin %
Trial Balance	ADSO on GL Totals	GL Account balance summary

Cost Centre Exp. Profit Centre P&L	CO-CCA InfoProvider EC-PCA CompositeProvider	Expense by cost centre × month Revenue vs Expense by PC
AR Ageing	FI-AR InfoProvider	Overdue receivables by bucket
Doc Type Audit	Compliance Query	Posting count by document type

SAP Databricks: *nb_period_close_pnl* — MoM variance + 3-month rolling revenue (simulates PySpark OVER window function). *nb_dso_scorecard* — Collection rate & avg overdue days per Profit Centre.

Datasphere Data Products (7 CSVs):
DP_GL_Journal_Entries, DP_PnL_By_Period, DP_Trial_Balance, DP_CostCentre_Expenses, DP_ProfitCentre_PnL, DP_DSO_Scorecard, DP_AR_Ageing — each published to the Datasphere Data Marketplace.

3. Tech Stack

Component	Technology	SAP BDC Equivalent
Language	Python 3.10+	SAP BDC Pipeline Scripts
OLAP Engine	DuckDB 0.10 (ANSI SQL)	SAP HANA/ BW/4HANA Bridge
Batch Processing	pandas + numpy	SAP Databricks(PySpark)
Visualization	Matpotlib + seaborn	SAP Analytics Cloud
Data Exports	CSV(pandas)	SAP Datasphere Data Marketplace
FI Concepts	GL, BKPF, BSEG, AR Ageing	SAP FI / CO Module
BW Concepts	CompositeProvider, ADSO, KFs	SAP BW/4HANA Bridge
Databricks	Window Func, GroupBy, MoM	SAP Databricks (Azure Runtime)
Version Control	GitHub (Public Repo)	SAP BDC DevOps

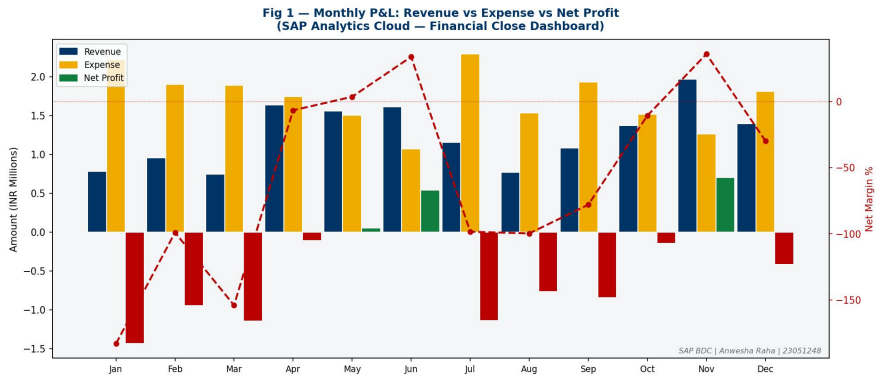
4. Pipeline Architechture

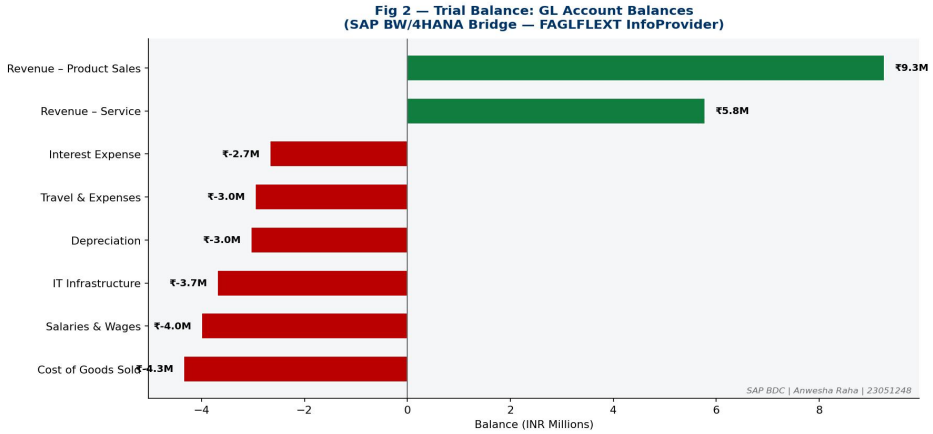
SAP S/4HANA (BKPF/BSEG/FAGLFLEXT) → **SAP Datasphere** (SDI/SLT ingestion) → **BW/4HANA Bridge** (6 OLAP queries: P&L, Trial Balance, Cost Centre, Profit Centre, AR Ageing, Audit) → **SAP Databricks** (nb_period_close_pnl | nb_dso_scorecard) → **SAP Analytics Cloud** (6 Financial Story Panels) → **Datasphere Data Products** (7 CSVs for downstream consumers such as SAP IBP and external auditors).

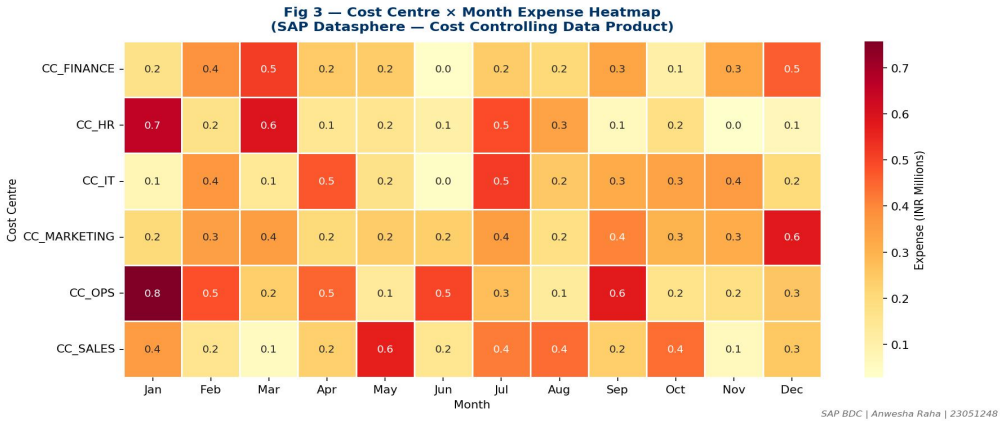
5. SAC Dashboard - 6 Financial Story Panels

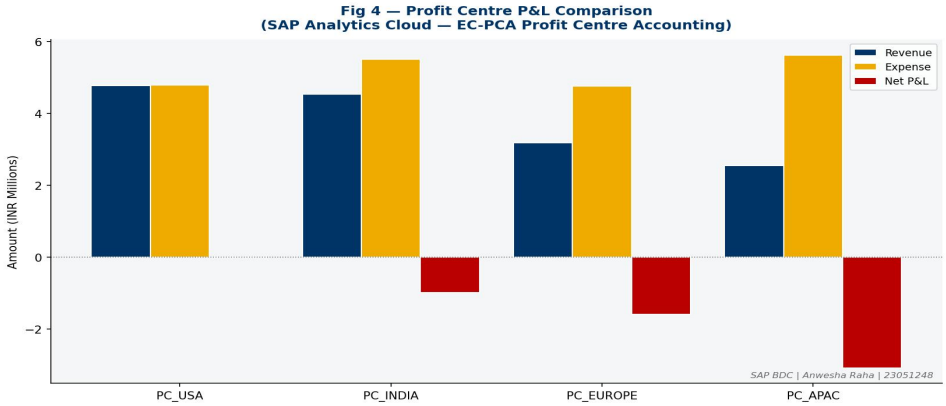
Figure	Chart Type	Financial Insight
Fig 1	Grouped Bar + Dual Axis	Monthly P& L: Revenuw vs Expense vs Net Profit + Margin %
Fig 2	Horizontal Bar	Trial Balance - GL Account Balances (BW/4HANA Bridge)
Fig 3	Heatmap	Cost Centre * Month Expense Heatmap
Fig 4	Grouped Bar	Profit Centre P&L Comparison
Fig 5	Grouped Bar + Bar	AR Ageing Buckets + Collection Rate
Fig 6	Pie + Line	Document Type Audit Distribution + Rolling Revenue Trend

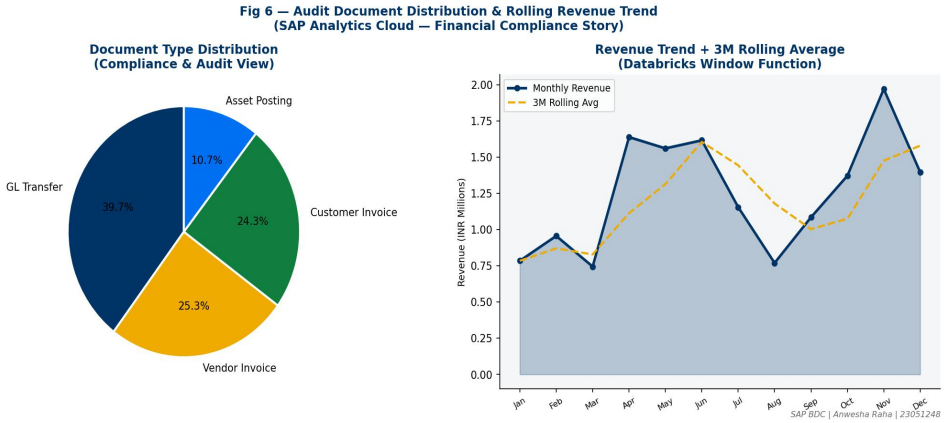
5. Screenshots











6. Unique Highlights

Complete R2R Coverage: All six R2R sub-processes (GL posting, P&L close, trial

balance, cost centre, profit centre, AR ageing) in one integrated BDC pipeline mirroring the
full financial controller FY 2024 workflow.

BW/4HANA Modernization:
CompositeProvider-style multi-dimension SQL aggregation with fiscal period characteristics aligned with BW/4HANA Bridge Clean Core best practices. Databricks Period Intelligence: MoM variance via pandas .shift() simulates PySpark LAG() window function; DSO scorecard with 80% collection rate target ,a standard R2R KPI.

Audit-Ready Compliance: Document type (SA/DR/KR/AB) distribution mirrors SAC FI audit story panels for internal and external year-end auditors.

SAP Brand-Compliant Visuals: All 6 charts use official SAP palette (#003366, #F0AB00, #107E3E) with student
watermarks, replicating real SAC financial exports without SAP GUI access.

7. Future Improvements

Enhancement	Description
Live S/4HANA OData V4	Connect to /API_GLACCOUNTLINEITEM_SRV for real GL journey entry ingestion
SAP Datasphere SDK	Publish 7 Data Products via Datasphere REST API /api/v1/dwc/catalog
Automated Period-Close	Databricks Job 9(1st of month) + CFO email alert on Net Profit threshold breach
Balance Sheet Generator	Auto generate Balance Sheet from GL account classification table
CDS View Migration	Migrate DuckDB queries to Datasphere CDS Views with @Analytics.dataCategory: #CUBE
Delta Lake Integration	Replace CSVexports with Delta Tables on ADLS Gen2 for incremental upserts.